

```
In [15]: import numpy as np
```

```
In [ ]: #Check link : https://www.youtube.com/watch?v=u-ym7BWHO-g&list=PLVLQHNRLfLP9vV2\_F08h1yiVD90XhJy8M&index=15
```

```
In [16]: ones_arr = np.ones((5,5))
ones_arr
```

```
Out[16]: array([[1., 1., 1., 1., 1.],
               [1., 1., 1., 1., 1.],
               [1., 1., 1., 1., 1.],
               [1., 1., 1., 1., 1.],
               [1., 1., 1., 1., 1.]])
```

```
In [17]: ones_arr = np.ones((5,5), dtype = int)
ones_arr
```

```
Out[17]: array([[1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1],
               [1, 1, 1, 1, 1]])
```

```
In [18]: zeros_arr = np.zeros((3,3), dtype = int)
zeros_arr
```

```
Out[18]: array([[0, 0, 0],
               [0, 0, 0],
               [0, 0, 0]])
```

```
In [19]: ones_arr * 255
```

```
Out[19]: array([[255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255],
               [255, 255, 255, 255, 255]])
```

```
In [20]: import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [21]: from PIL import Image
```

```
In [28]: my_img = Image.open(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Practice\labubu.jpg')  
my_img
```

Out[28]:





```
In [29]: type(my_img)
```

```
Out[29]: PIL.WebPImagePlugin.WebPImageFile
```

```
In [30]: img_arr = np.asarray(my_img)
```

```
In [31]: img_arr
```

```
Out[31]: array([[190, 230, 239],
               [190, 230, 239],
               [190, 230, 239],
               ...,
               [174, 187, 217],
               [174, 187, 217],
               [172, 188, 217]]],

          [[190, 230, 239],
           [190, 230, 239],
           [190, 230, 239],
           ...,
           [175, 186, 217],
           [174, 187, 217],
           [174, 187, 217]]],

          [[190, 230, 239],
           [190, 230, 239],
           [190, 230, 239],
           ...,
           [177, 185, 217],
           [175, 186, 217],
           [175, 186, 217]]],

          ...,

          [[200, 159, 107],
           [200, 159, 107],
           [200, 159, 107],
           ...,
           [182, 180, 76],
           [183, 182, 77],
           [183, 182, 77]]],

          [[198, 158, 98],
           [198, 158, 98],
           [198, 158, 98],
           ...,
           [183, 180, 76],
           [182, 180, 76],
           [182, 180, 76]]],
```

```
[[196, 156, 92],
 [196, 156, 92],
 [196, 156, 92],
 ...,
 [183, 180, 76],
 [182, 180, 76],
 [182, 180, 76]]], dtype=uint8)
```

```
In [32]: type(img_arr)
```

```
Out[32]: numpy.ndarray
```

```
In [38]: img_arr.shape
```

```
Out[38]: (823, 823, 3)
```

```
In [33]: plt.imshow(img_arr)
```

```
Out[33]: <matplotlib.image.AxesImage at 0x1ad240d74d0>
```

```
In [34]: plt.show()
```



```
In [35]: my_img_red = img_arr.copy()  
my_img_red
```

```
Out[35]: array([[190, 230, 239],
               [190, 230, 239],
               [190, 230, 239],
               ...,
               [174, 187, 217],
               [174, 187, 217],
               [172, 188, 217]]],

          [[190, 230, 239],
           [190, 230, 239],
           [190, 230, 239],
           ...,
           [175, 186, 217],
           [174, 187, 217],
           [174, 187, 217]]],

          [[190, 230, 239],
           [190, 230, 239],
           [190, 230, 239],
           ...,
           [177, 185, 217],
           [175, 186, 217],
           [175, 186, 217]]],

          ...,

          [[200, 159, 107],
           [200, 159, 107],
           [200, 159, 107],
           ...,
           [182, 180, 76],
           [183, 182, 77],
           [183, 182, 77]]],

          [[198, 158, 98],
           [198, 158, 98],
           [198, 158, 98],
           ...,
           [183, 180, 76],
           [182, 180, 76],
           [182, 180, 76]]],
```



```
[[196, 156, 92],  
 [196, 156, 92],  
 [196, 156, 92],  
 ...,  
 [183, 180, 76],  
 [182, 180, 76],  
 [182, 180, 76]]], dtype=uint8)
```

```
In [39]: img_arr == my_img_red
```

```
Out[39]: array([[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

               [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

               [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

               ...,

               [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]],

               [[ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True],
               ...,
               [ True,  True,  True],
               [ True,  True,  True],
               [ True,  True,  True]]]
```

```
[[ True,  True,  True],  
 [ True,  True,  True],  
 [ True,  True,  True],  
 ...,  
 [ True,  True,  True],  
 [ True,  True,  True],  
 [ True,  True,  True]])
```

```
In [40]: plt.imshow(my_img_red)
```

```
Out[40]: <matplotlib.image.AxesImage at 0x1ad25691a90>
```

```
In [41]: plt.show()
```



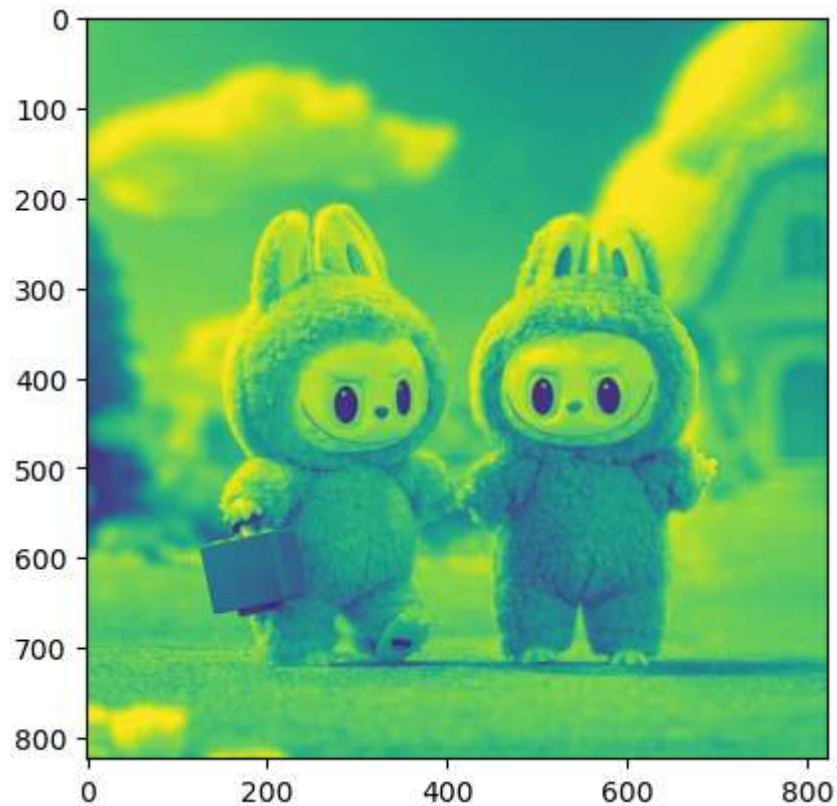
```
In [42]: my_img_red.shape
```

```
Out[42]: (823, 823, 3)
```

```
In [43]: plt.imshow(my_img_red[:, :, 0])
```

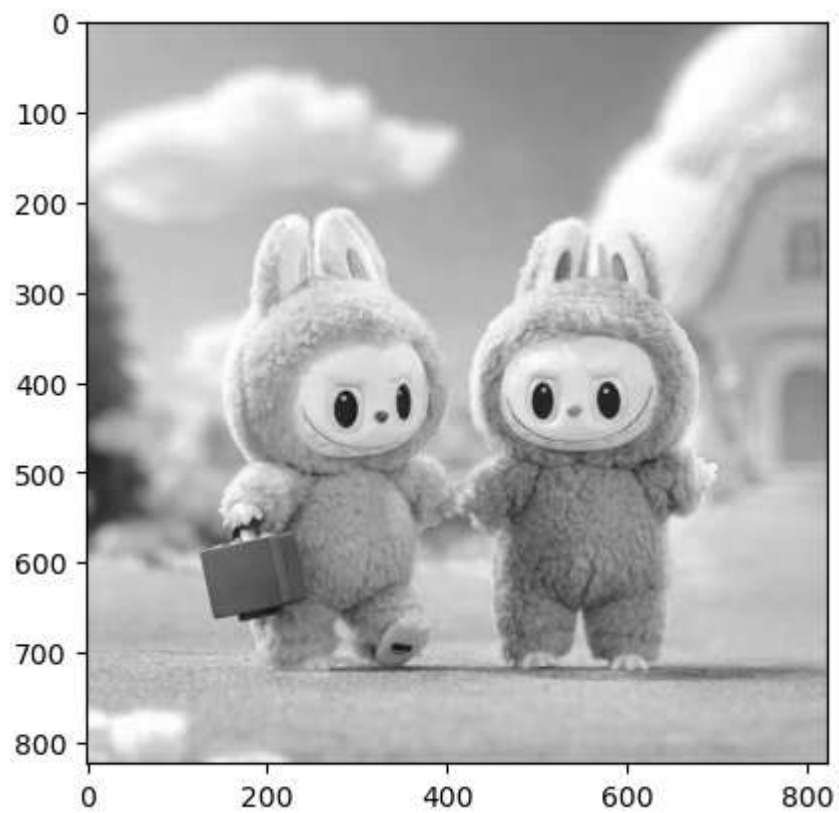
```
Out[43]: <matplotlib.image.AxesImage at 0x1ad2570f750>
```

```
In [44]: plt.show()
```

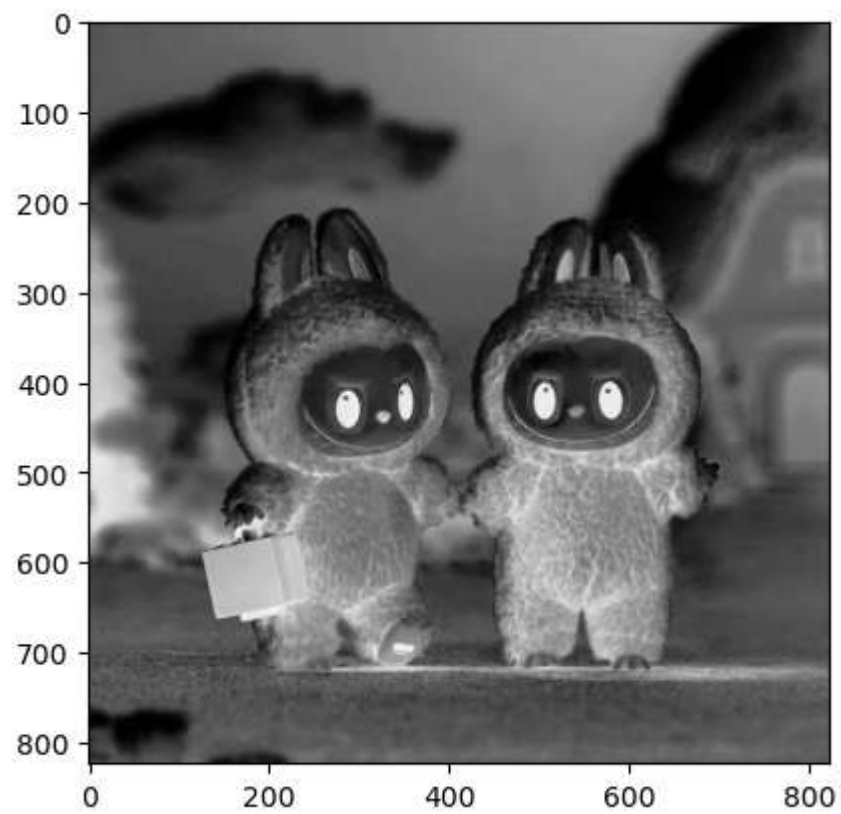


```
In [ ]: # Refer this website https://matplotlib.org/stable/gallery/color/colormap\_reference.html
```

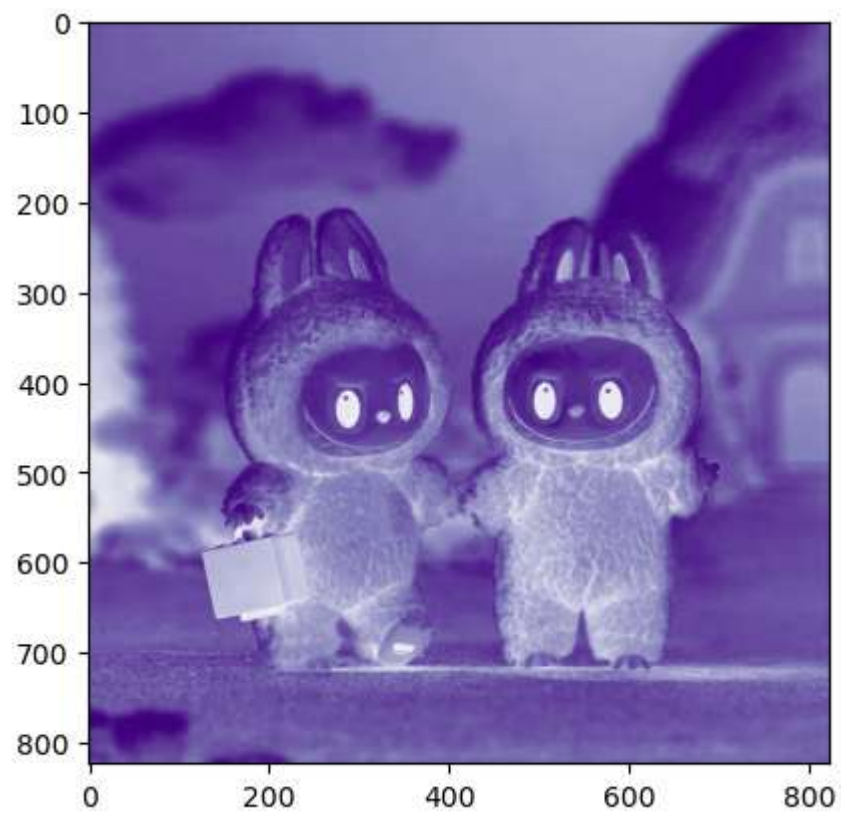
```
In [45]: plt.imshow(my_img_red[:, :, 0], cmap = 'grey')  
plt.show()
```



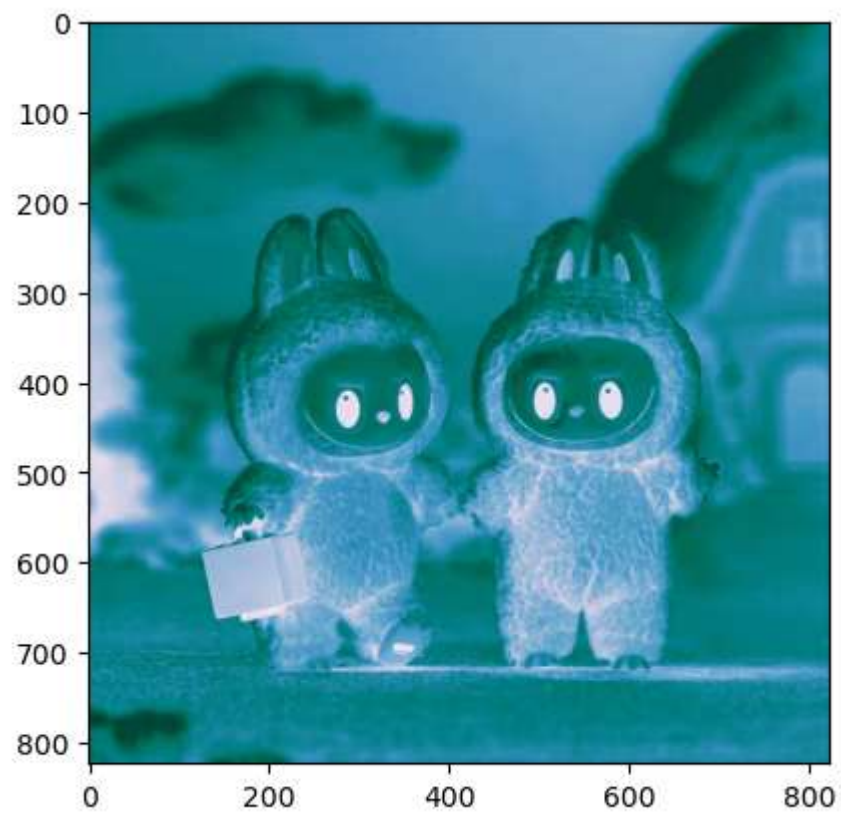
```
In [46]: plt.imshow(my_img_red[:, :, 0], cmap = 'Greys')  
plt.show()
```



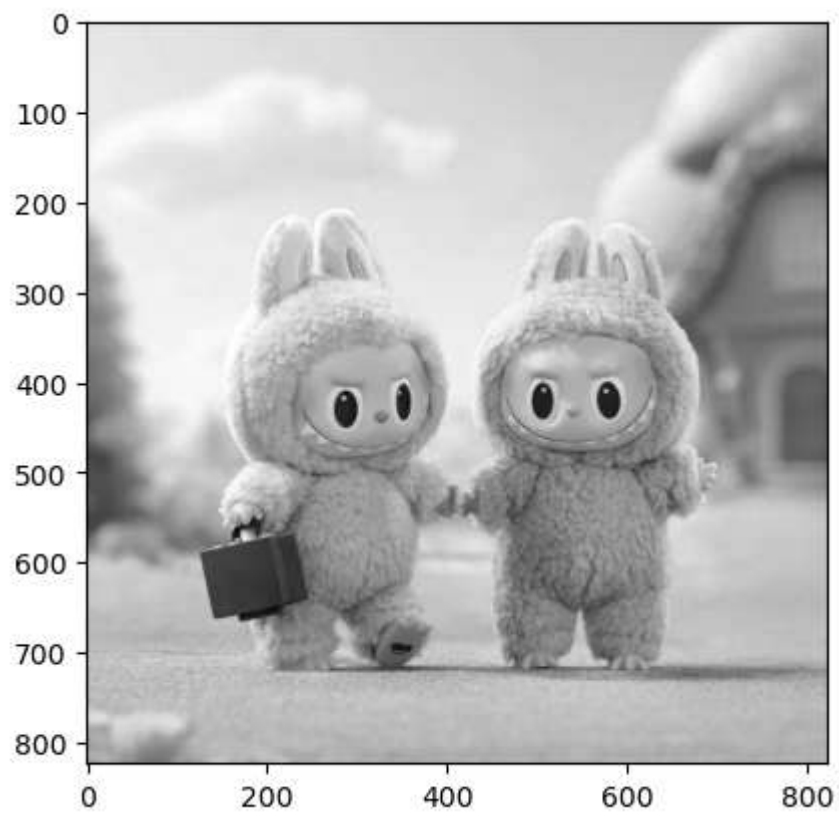
```
In [47]: plt.imshow(my_img_red[:, :, 0], cmap = 'Purples')  
plt.show()
```



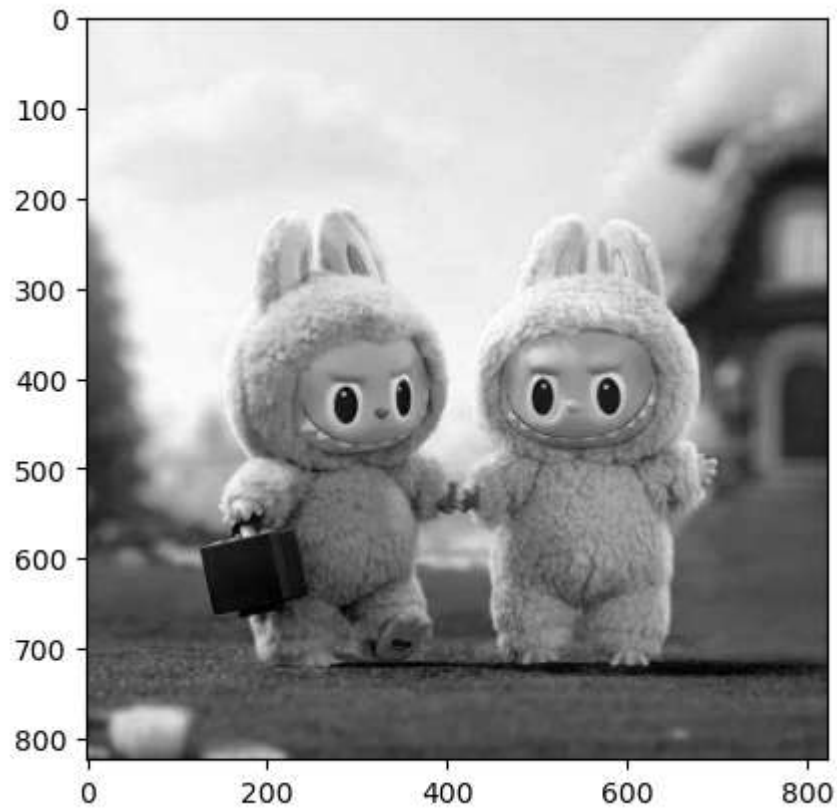
```
In [48]: plt.imshow(my_img_red[:, :, 0], cmap = 'PuBuGn')  
plt.show()
```



```
In [49]: plt.imshow(my_img_red[:, :, 1], cmap = 'grey')  
plt.show()
```

```
In [50]: plt.imshow(my_img_red[:, :, 2], cmap = 'grey')  
plt.show()
```



```
In [51]: my_img_red[:, :, 0]
```

```
Out[51]: array([[190, 190, 190, ..., 174, 174, 172],
                [190, 190, 190, ..., 175, 174, 174],
                [190, 190, 190, ..., 177, 175, 175],
                ...,
                [200, 200, 200, ..., 182, 183, 183],
                [198, 198, 198, ..., 183, 182, 182],
                [196, 196, 196, ..., 183, 182, 182]], dtype=uint8)
```

```
In [52]: my_img_red[:, :, 1]
```

```
Out[52]: array([[230, 230, 230, ..., 187, 187, 188],
               [230, 230, 230, ..., 186, 187, 187],
               [230, 230, 230, ..., 185, 186, 186],
               ...,
               [159, 159, 159, ..., 180, 182, 182],
               [158, 158, 158, ..., 180, 180, 180],
               [156, 156, 156, ..., 180, 180, 180]], dtype=uint8)
```

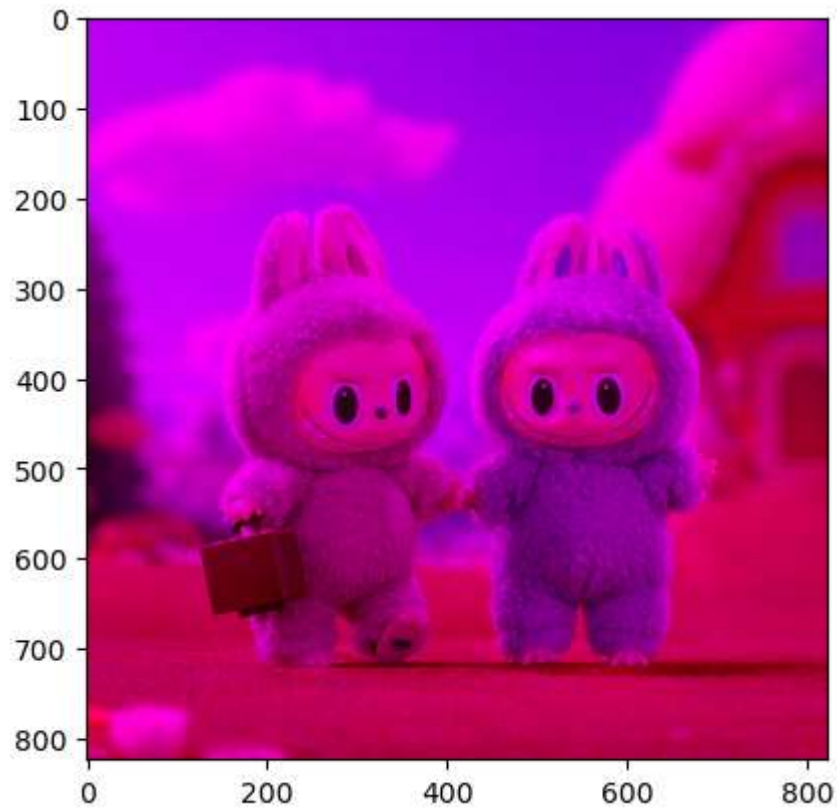
```
In [53]: my_img_red[:, :, 2]
```

```
Out[53]: array([[239, 239, 239, ..., 217, 217, 217],
               [239, 239, 239, ..., 217, 217, 217],
               [239, 239, 239, ..., 217, 217, 217],
               ...,
               [107, 107, 107, ..., 76, 77, 77],
               [ 98, 98, 98, ..., 76, 76, 76],
               [ 92, 92, 92, ..., 76, 76, 76]], dtype=uint8)
```

```
In [54]: my_img_red[:, :, 1] = 0
         my_img_red[:, :, 1]
```

```
Out[54]: array([[0, 0, 0, ..., 0, 0, 0],
               [0, 0, 0, ..., 0, 0, 0],
               [0, 0, 0, ..., 0, 0, 0],
               ...,
               [0, 0, 0, ..., 0, 0, 0],
               [0, 0, 0, ..., 0, 0, 0],
               [0, 0, 0, ..., 0, 0, 0]], dtype=uint8)
```

```
In [55]: plt.imshow(my_img_red)
         plt.show()
```



```
In [56]: my_img_red[:, :, 2] = 0  
my_img_red[:, :, 2]
```

```
Out[56]: array([[0, 0, 0, ..., 0, 0, 0],  
                [0, 0, 0, ..., 0, 0, 0],  
                [0, 0, 0, ..., 0, 0, 0],  
                ...,  
                [0, 0, 0, ..., 0, 0, 0],  
                [0, 0, 0, ..., 0, 0, 0],  
                [0, 0, 0, ..., 0, 0, 0]], dtype=uint8)
```

```
In [57]: plt.imshow(my_img_red)  
plt.show()
```



```
In [61]: arr1 = np.asarray(my_img)
arr1
```

```
Out[61]: array([[190, 230, 239],
               [190, 230, 239],
               [190, 230, 239],
               ...,
               [174, 187, 217],
               [174, 187, 217],
               [172, 188, 217]]],

          [[190, 230, 239],
           [190, 230, 239],
           [190, 230, 239],
           ...,
           [175, 186, 217],
           [174, 187, 217],
           [174, 187, 217]]],

          [[190, 230, 239],
           [190, 230, 239],
           [190, 230, 239],
           ...,
           [177, 185, 217],
           [175, 186, 217],
           [175, 186, 217]]],

          ...,

          [[200, 159, 107],
           [200, 159, 107],
           [200, 159, 107],
           ...,
           [182, 180, 76],
           [183, 182, 77],
           [183, 182, 77]]],

          [[198, 158, 98],
           [198, 158, 98],
           [198, 158, 98],
           ...,
           [183, 180, 76],
           [182, 180, 76],
           [182, 180, 76]]],
```

```
[[196, 156, 92],  
 [196, 156, 92],  
 [196, 156, 92],  
 ...,  
 [183, 180, 76],  
 [182, 180, 76],  
 [182, 180, 76]]], dtype=uint8)
```

```
In [62]: type(arr1)
```

```
Out[62]: numpy.ndarray
```

```
In [63]: arr1.shape
```

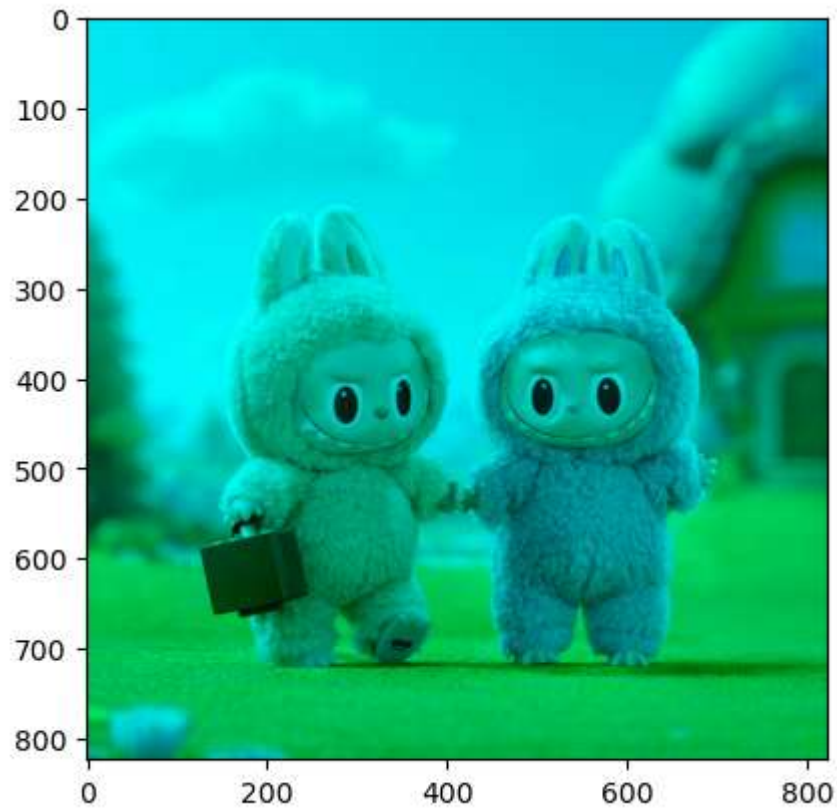
```
Out[63]: (823, 823, 3)
```

```
In [65]: plt.imshow(arr1)  
plt.show()
```



```
In [66]: my_img1 = arr1.copy()
         my_img1[:, :, 0] = 0
```

```
In [67]: plt.imshow(my_img1)
         plt.show()
```

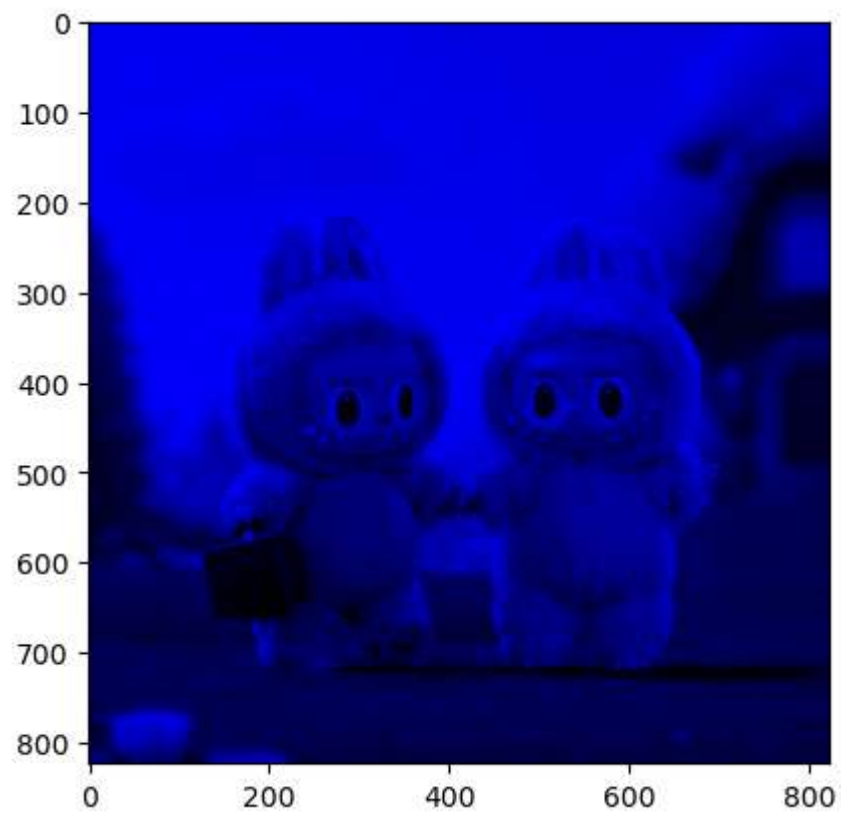



```
In [68]: my_img1[:, :, 1]
```

```
Out[68]: array([[230, 230, 230, ..., 187, 187, 188],
                [230, 230, 230, ..., 186, 187, 187],
                [230, 230, 230, ..., 185, 186, 186],
                ...,
                [159, 159, 159, ..., 180, 182, 182],
                [158, 158, 158, ..., 180, 180, 180],
                [156, 156, 156, ..., 180, 180, 180]], dtype=uint8)
```

```
In [69]: my_img1[:, :, 1] = 0
```

```
In [71]: plt.imshow(my_img1)
plt.show()
```



In []: